

Project ID:

R26-SE-011

1. Topic (12 words max)

Blockchain-Based Transparent and Secure Academic Grading Using Decentralized Verification

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature & Gravity of the Problem

Educational institutions currently rely on centralized databases (SQL) to store academic records. These centralized "silos" are vulnerable to single points of failure, including internal manipulation by corrupt administrators and external cyberattacks. This vulnerability critically affects the **student** (User Persona), whose credentials risk devaluation due to rampant diploma fraud, and **Employers**, who struggle with slow, expensive manual verification processes.

Impact of the Problem

- **Technical Impact:** Current systems lack an immutable audit trail; once a grade is altered in a standard database, the history is often lost. Furthermore, existing blockchain pilots suffer from the "Oracle Problem" (Garbage-In, Garbage-Out), where incorrect data is uploaded without validation.
- **Economic & Ethical Impact:** Degree mills and credential fraud cost the global economy billions annually and erode trust in higher education [1]. Ethically, students face a "black box" scenario where they cannot verify if their finalized grades were processed fairly or altered post-submission.

Sustainable Development Goals (SDGs)

This project directly addresses **SDG 4 (Quality Education)** by preserving the integrity of qualifications and ensuring equitable access to verifiable credentials. It also supports **SDG 16 (Peace, Justice, and Strong Institutions)** by reducing corruption and fraud through transparent, tamper-proof institutional record-keeping [2].

References

- [1] R. F. Cheng, "Credential Fraud and the consequent costs to the Global Economy," *Journal of Academic Ethics*, vol. 18, no. 2, pp. 112-125, 2022.
- [2] A. Grech and A. F. Camilleri, "Blockchain in Education," *JRC Science for Policy Report, European Commission*, 2017.
- [3] N. M. K. Al-Zoubi et al., "A Survey on Blockchain for Smart Education: Security and Privacy," *IEEE Access*, vol. 9, pp. 99-115, 2021.

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products / Systems

Most universities today use centralized Learning Management Systems (LMS) such as Moodle, Canvas, or Blackboard to manage student assessments and grades. In these systems, grades are stored in centralized databases controlled by lecturers or administrators. While these platforms are easy to use, they rely heavily on a single authority, making them vulnerable to grade manipulation, data loss, lack of transparency, and limited independent verification. Students and external parties such as employers cannot easily verify whether grades are authentic or unchanged over time.

Summary of Existing Research

Several studies have explored the use of blockchain technology in education to improve transparency, security, and trust. For example, Ayasrah et al. proposed a blockchain-based student assessment system that uses smart contracts to automate grading processes and store assessment records in an immutable ledger. This approach ensures that grades cannot be altered once recorded and allows stakeholders to verify assessment outcomes independently.

Other research highlights blockchain's ability to prevent data tampering, improve record integrity, and enable verifiable digital certificates. Existing academic solutions mainly focus on secure record storage, credential verification, and automation of assessments, often using Ethereum-based smart contracts.

Gaps and Limitations Identified

Despite these advancements, current research and systems have several limitations. Most proposed solutions do not address subjective grading, such as essays or projects, where human judgment is required. Many systems still rely on single-author grading, which can introduce bias. Scalability and transaction cost issues remain unresolved, especially when handling large numbers of students. Additionally, privacy concerns exist because blockchain immutability can conflict with data protection regulations. Integration with existing LMS platforms is also limited, creating high adoption barriers for universities. These gaps highlight the need for a scalable, privacy-aware, multi-reviewer, and LMS-integrated blockchain grading system, which this research aims to address.

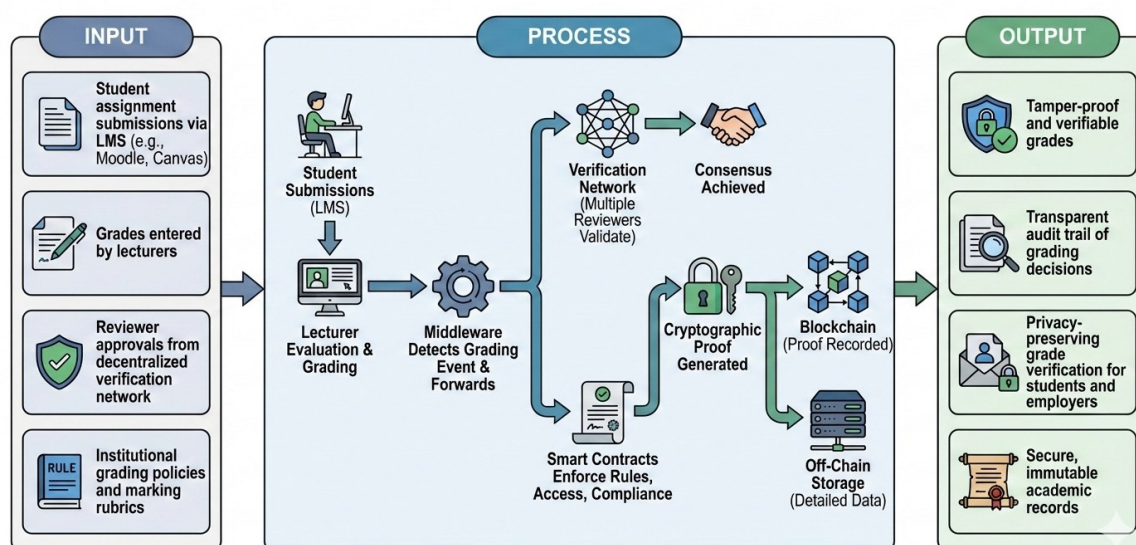
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the Identified Gaps

The proposed solution is a Blockchain-Based Transparent and Secure Grading System designed to overcome the limitations of existing centralized grading systems and prior blockchain-based research. Unlike traditional LMS platforms that rely on a single authority, this system introduces decentralized grade verification, ensuring fairness and reducing bias. It directly addresses subjective grading challenges by using a multi-reviewer verification network, where grades are approved through consensus rather than individual judgment.

Scalability and cost issues identified in earlier research are resolved through a Layered Blockchain Architecture, where only cryptographic proofs are stored on the main blockchain, while large files and personal data are stored securely off-chain. Privacy conflicts with regulations such as GDPR and FERPA are handled using Zero-Knowledge Proofs (ZKPs), allowing grades to be verified without exposing student identities. Furthermore, seamless integration with existing LMS platforms ensures low adoption barriers, allowing institutions to adopt the system without replacing their current infrastructure.

System Perspective:



Nature of the Solution

Overall, the proposed system is decentralized, secure, privacy-aware, and scalable, combining blockchain technology, smart contracts, and seamless LMS integration. It enhances trust in academic grading by ensuring fairness, transparency, and long-term verifiability, while remaining practical for real-world university environments.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Availability of Specialized Domain Expertise

The successful implementation of the proposed blockchain-based grading system requires expertise from multiple domains, including software engineering, blockchain technology, cybersecurity, and educational assessment systems. Members of the project team can gain essential domain expertise through a combination of academic coursework, supervised research, and self-directed learning. Relevant university modules such as Distributed Systems, Software Security, Blockchain Technologies, and Software Architecture provide a strong theoretical foundation. Additionally, consulting academic research papers, industry whitepapers, and blockchain documentation enables team members to understand current best practices and limitations in blockchain-based educational systems.

Domain expertise related to academic grading policies, assessment workflows, and LMS usage can be gained through collaboration with lecturers, teaching assistants, and academic administrators. Reviewing institutional grading guidelines and assessment rubrics further supports accurate system design aligned with real-world academic processes.

Acquisition of Technical Knowledge and Skills

The technical knowledge required for this project includes smart contract development, decentralized system design, cryptographic techniques, and secure web application integration. Team members can acquire these skills through online learning platforms, official blockchain documentation, and hands-on experimentation. Technologies such as Ethereum, Solidity, Layer-2 solutions, IPFS, and Zero-Knowledge Proof frameworks can be learned using open-source tutorials and developer communities.

Practical experience can be gained by developing small prototypes, participating in blockchain test networks, and using version control systems such as GitHub for collaborative development. Familiarity with LMS APIs (e.g., Moodle or Canvas) can be achieved through documentation review and API testing in controlled environments.

Data Collection for the Project

The project primarily uses non-sensitive, simulated academic data for development and testing purposes. Sample datasets include anonymized student profiles, mock assignment submissions, grading records, and reviewer approvals. These datasets can be generated synthetically or derived from publicly available educational data formats. No real student academic records are required during the research and prototyping stages. System logs, transaction metadata, and performance metrics are collected automatically during system execution to evaluate efficiency, scalability, and correctness.

Ethical Clearance Considerations

Since the project does not involve real student identities, personal academic records, or human subject experimentation, formal ethical clearance is not required at the current research and prototype level. However, ethical principles such as data minimization, privacy protection, and informed consent are incorporated into the system design. If the system is later deployed in a real institutional environment using actual student data, approval from the relevant ethical review board or institutional authority would be required to ensure compliance with data protection and privacy regulations.

9. Objectives and Novelty

Main Objective Out main objective is o design and develop a decentralized, privacy-preserving academic grading framework that integrates with existing Learning Management Systems (LMS) to ensure immutable record-keeping, verifiable transparency, and fair subjective evaluation through a validator consensus network.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera M.A.I - IT22206282	System Architecture (L1 + IPFS + ZKPs)	1. Design and deploy smart contracts on a Layer-1 blockchain to store cryptographic hashes of grading records. 2. Integrate InterPlanetary File System (IPFS) for off-chain storage of assessment files and grading metadata to reduce on-chain storage costs. 3. Implement Zero-Knowledge Proof (ZKP) mechanisms to enable grade verification without revealing student identity or raw grade data.	Privacy-Preserving Grade Integrity: Unlike conventional blockchain-based academic systems that store grades or student details directly on-chain, this component combines off-chain storage with zero-knowledge proofs to ensure tamper-proof grade verification while fully preserving student privacy.

Dilki H P C - IT22111210	Decentralized Verification Network (DON)	1. Design a multi-reviewer grade validation workflow. 2. Implement threshold-based consensus logic for grade approval. 3. Develop incentive and penalty mechanisms to encourage honest reviewer behavior and discourage collusion or malicious approvals.	Decentralized Subjective Grade Verification: Unlike existing blockchain-based grading systems that rely on single-teacher decisions, this component introduces a consensus-driven verification network for subjective assessments, improving fairness and trust in academic grading.
Perera W A N I - IT22061348	Seamless LMS Integration	1. Develop API connectors to fetch data from LMS webhooks. 2. Build the "listener" service that triggers blockchain transactions automatically. 3. Develop the public Verification Portal for employers.	Frictionless Adoption: Introduces a "silent" blockchain layer that requires no behavioral change from lecturers, solving the high barrier to entry in existing EdTech solutions.

Perera N S G - IT22276346	Security & Formal Verification	1. Formally verify smart contract logic using Certora or TLA+ to prove the absence of logic flaws and state-manipulation vulnerabilities mathematically. 2. Conduct security audits by simulating Sybil and collusion threat models to ensure the resilience of the validator consensus network. 3. Optimize gas consumption through code refactoring and profiling to minimize transaction costs for high-volume academic record processing.	Formally Verified Integrity: Unlike standard educational platforms, this framework utilizes formal verification to provide a mathematical guarantee of correctness. By moving beyond traditional testing, it ensures that the grading logic is immune to the state-manipulation bugs that frequently plague decentralized systems..
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**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	
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- b) The proposed topic exhibits novelty.

Yes	✓	No	
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- c) The student group can successfully execute the proposed project.

Yes	✓	No	
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	
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- e) Any other comments:

Recommend proceeding with this project.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Junius Anjana	Vidanaralage	
Co-Supervisor	Ms.	Thisara	Shyamalee	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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